

MAFE Financial Econometrics

Maximum Likelihood Estimation (MLE)

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MLE - Learning Road Map

1. Lecture:

- An introduction to MLE with a Poisson distribution example.

2. Pre-class take-home exercise:

- Solidify your understanding of the lecture material in preparation for the applied class exercise.

3. Applied class exercise:

- Estimating market return and risk from S&P 500 data.

Teaching vision

- From technique to judgement
- Question $\rightarrow$ method $\rightarrow$ implementation $\rightarrow$ interpretation
- Structured rigour: data, code, diagnosis
- Inclusive access without dilution
- Future: applied quantitative learning design

Administration

- Teaching quality needs infrastructure
- Coherent modules; clear expectations
- Tutor coordination; shared standards
- Assessment fairness; moderation
- Student difficulty → targeted support

Academic citizenship

- Beyond my own classroom
- Personal tutoring as academic navigation
- Mentoring GTAs and early-career teachers
- Widening participation; realistic pathways
- Shared practice, not isolated excellence

Thank you!